

KUSHAL SINGH

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Education

Chandigarh University
B.E. in Computer Science
CGPA: 7.94

Punjab, India
2022 – 2026

Experience

Indian Institute of Technology (IIT BHU)

Jun 2025 – Jul 2025

Research Intern

- Conducted research on IoT network traffic classification using bit-level binary data and machine learning.
- Developed lightweight models using Random Forest and CNN for IoT device identification using only first 128–256 bits of packet payload.
- Proposed efficient traffic analysis methods for low-power IoT devices.

WhatBytes

Mar 2025 – Apr 2025

Backend Developer Intern

- Built backend services using Python and Django with REST APIs and JWT authentication.
- Integrated payment gateways and implemented secure login/email authentication.
- Automated background tasks using Celery and implemented data scraping pipelines.

Vikas Parishad NGO

May 2024 – Jul 2024

Frontend Web Developer Trainee

- Developed responsive UI using HTML, CSS, JavaScript and Bootstrap.
- Improved NGO website performance and accessibility.

Projects

Healthcare Patient Data Analysis

Python, Pandas, SQLite, Matplotlib, Seaborn

- Analyzed 55,500+ real-world patient records using Python and SQLite to extract clinical and billing insights.
- Wrote 6 SQL queries using GROUP BY, AVG, COUNT and ORDER BY to identify disease-wise patient distribution and average billing trends.
- Built 4 data visualizations using Matplotlib and Seaborn covering gender distribution, test results and treatment costs.

Two-Wheeler Market Analysis – India (2025)

Python, Pandas, Matplotlib, MS Excel

- Analyzed two-wheeler sales data across 4 Indian cities (Bhopal, Chandigarh, Noida, Bengaluru) covering 4 major brands over 3 months.
- Identified consumer spending patterns by grouping model-wise sales into price bands (Rs.80k–1L dominant segment across all cities).
- Discovered festive season impact – November showed 20–30% spike in mid-range (Rs.1.2L–1.4L) purchases compared to Jan–Feb.
- Concluded Bengaluru leads in EV adoption while Bhopal shows highest price sensitivity among analyzed markets.

Technical Skills

Languages: Python, SQL, C

Data Tools: Pandas, NumPy, Excel, Matplotlib, Seaborn

Frameworks: Django, REST Framework

Tools: Git, Kafka, SQLite, REST APIs

Concepts: Data Analysis, Machine Learning, OOP, Data Structures & Algorithms

Publications

- Integrating Gene Expression and Imaging Data for Improved Cancer Diagnosis – IJEDR
- Unveiling the Power of Personalization in Health and Fitness Apps – IJRASET